An Adaptive Explainable Cognitive Graph Digital Twin for Cyber-Resilient Smart Hospital Energy Ecosystems

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Abstract. Smart hospital campuses rely on localized microgrids to sustain intensive-care ventilators, surgical robotics, and biomedical cold chains. A stealthy False Data Injection Attack (FDIA) on substation Remote Terminal Units can corrupt the energy-management state estimator while evading its χ^2 residual check, placing patient-critical loads at risk. We introduce a Cognitive Graph Digital Twin (CGDT) that mirrors the hospital electrical topology as a dynamic admittance graph and runs a multi-task Admittance-Conditioned GraphSAGE backbone for simultaneous attack detection and bus-level localization in a single forward pass at 0.908 ms latency. To bridge the gap between post-hoc saliency diagnostics and inductive training, our Attribution-Guided Multi-Task Loss supervises backpropagation gradients via attribution alignment to target true substation anomaly locations, driving a 5.52 percentage-point improvement in localization F1. Dynamic physical-to-virtual coherence is governed by a state-space tracking filter coupled with an adaptive exponential baseline. Upon detecting an attack, a closed-loop healing module quarantines compromised sensor channels and reconstructs measurements via graph-guided spatial interpolation, reducing Weighted Least Squares estimation error by 99.97%. On the IEEE 14-bus test feeder under full nonlinear AC power flow, the CGDT achieves $98.25\% \pm 0.34\%$ detection F1 and $97.62\% \pm 0.84\%$ localization F1 across five random seeds, and retains 92.80% detection F1 under zero-shot topology reconfigurations without retraining.

Keywords: Graph Digital Twin · Healthcare Cybersecurity · GraphSAGE · Explainable AI · False Data Injection Attacks · Smart Hospital Microgrid

1 Introduction

Hospital distribution microgrids combine utility feeds with rooftop PV, battery storage, and diesel generation to supply clinical corridors [6,17]. In surgical suites

and intensive-care units (ICUs), power quality is non-negotiable: sub-cycle sags stall robotic surgical tools, ventilators require continuous supply, and pharmacy cold chains require uninterrupted −80°C chilling. Under IEC 60601-1, nuisance trips or false load shedding directly threaten patient life.

Within substation automation (IEC 61850), Remote Terminal Units report voltage magnitudes, angles, and power flows $z = [V, \theta, P, Q]^T \in \mathbb{R}^m$ to energy management systems. Conventional state estimators apply Weighted Least Squares (WLS) to determine states $\hat{x} \in \mathbb{R}^{2N-1}$, validating measurement consistency through χ^2-residual gating: $J(\hat{x}) = \|z - h(\hat{x})\|^2_{R^{-1}} \leq \tau_{\chi^2}$. Coordinated False Data Injection Attacks (FDIAs) undermine this residual check. Given full or partial topology access, an attacker evaluates the measurement Jacobian $H = \partial h(\hat{x})/\partial x$ and injects $a = Hc$. Because $h(\hat{x} + c) \approx h(\hat{x}) + Hc$, the perturbed measurement vector $z_a = z + a$ yields an invariant residual $\|z_a - h(\hat{x} + c)\|^2_{R^{-1}} \equiv \|z - h(\hat{x})\|^2_{R^{-1}} \leq \tau_{\chi^2}$, misleading automated voltage regulators and tie-line switches without raising SCADA bad-data alarms [15,14].

Early machine-learning defenses deployed static classifiers—specifically support vector machines, multi-layer perceptrons, and 1D-CNNs [10,3]—operating on fixed-dimension feature vectors $z \in \mathbb{R}^m$. In hospital distribution topologies, however, routine operational actions (breaker switching, feeder transfers, scheduled maintenance) alter the admittance graph $G = (V, E)$. Once line connectivity switches, fixed measurement matrices lose channel alignment, causing severe classification breakdown unless models are fully retrained—a fatal limitation against 2–4 s SCADA telemetry polling.

Graph neural networks (GNNs) overcome topological rigidity by mapping substations to node sets V and physical feeders to edge sets E [21]. Breaker operations simply update adjacency elements $A_{uv} \in \{0, 1\}$, preserving node feature invariance. Nevertheless, current grid GNN formulations—including ARMA wavelets [4] and isotropic spatial convolutions [22]—treat transmission branches as binary or unweighted links. By omitting complex branch admittances $Y_{uv} = G_{uv} + jB_{uv}$, isotropic aggregation acts as an unweighted low-pass filter, blurring sharp voltage sags and localized phase distortions across low-impedance feeder corridors.

Real-time deployment in clinical environments also imposes strict latency and interpretability constraints. Traditional explainability frameworks (SHAP, LIME) evaluate hundreds of Monte Carlo perturbations, taking far too long for sub-second SCADA loops [27]. Furthermore, detecting anomalies without restoring telemetry fails to protect downstream controllers from diverging. Securing healthcare microgrids requires joint, millisecond-scale attack localization and autonomous state reconstruction before protection relays trip.

1.1 Key Contributions

To address these vulnerabilities, we introduce an explainable Cognitive Graph Digital Twin (CGDT) engineered for hospital microgrid infrastructure:

– **Admittance-Conditioned GraphSAGE**: By embedding line admittances $Y_{uv} = G_{uv} + jB_{uv}$ directly into message aggregation and spatial pooling, our

model prevents isotropic smoothing and captures steep voltage drops across hospital feeders.
- **State-Space Twin Tracking**: Coupling steady-state Kalman filtering with an exponential reference baseline ($\beta = 0.95$) decouples diurnal load dynamics from stealthy adversarial offsets, preventing a 16.32 pp collapse in localization F1.
- **Attribution-Guided Multi-Task Training (AG-MTL)**: Integrating 10-step Integrated Gradients ($S = 10$) into the loss function penalizes attribution leakage into unattacked buses during backpropagation, yielding a 5.52 pp improvement in bus localization precision.
- **Autonomous Sub-2 ms State Healing**: Following attack localization, an admittance-weighted spatial estimator interpolates corrupted telemetry from trusted neighbors, maintaining the full column rank of Jacobian H and recovering 99.97% of WLS state accuracy.

2 Related Work

2.1 Threat Landscape in Cyber-Physical Healthcare Microgrids

Modern medical centers deploy multi-tier IT-OT architectures wherein supervisory SCADA links interface with localized microgrids (photovoltaics, battery storage, backup diesel) to sustain life-support infrastructure under IEC 60601-1 and IEEE 1547 standards [13,6]. Corrupting telemetry along clinical feeders risks triggering unwarranted breaker trips or false tap changes, directly compromising intensive care units and surgical operating suites [17]. Prior healthcare security research addressed cyber risk via Bayesian network modeling [5] and medical IoT perimeter monitoring [18]. However, these approaches treat sensor streams as decoupled scalar channels, ignoring underlying circuit admittances Y_{bus} and physical AC power flow constraints.

2.2 State Estimation and False Data Injection Defense

Since Liu et al. [15] formalized the algebraic stealth criterion $a = Hc$ against linear DC state estimation, stealthy injection attacks have been extended to full AC power flow regimes, distributed microgrids, and partially observable networks [14,7], as well as DC microgrid systems [3,19]. To counter these attacks, tabular deep neural networks [10] and recurrent neural architectures [1] were introduced. However, because tabular architectures map each sensor channel to fixed input indices, topology changes (e.g., line isolation or bus tie-switching) corrupt feature alignment and trigger persistent false alarms.

2.3 Graph Learning and Explainability in Power Systems

To handle dynamic switching, attributed graph representations $G = (V, E)$ map substation buses to nodes and transmission branches to edges [21]. While spectral

methods [12], graph attention [25], and inductive GraphSAGE convolutions [9] have been evaluated for power networks [4,22,28,16], existing architectures overlook complex line admittances ($Y_{uv} = G_{uv} + jB_{uv}$), causing isotropic message smoothing across high-gradient electrical boundaries. Furthermore, post-hoc explainers such as SHAP or GNNExplainer require 10^2–10^3 Monte Carlo perturbations [27,26], exceeding sub-second SCADA loops. Concurrently, digital twin paradigms [8,23] have demonstrated significant value in industrial monitoring [2] and healthcare facilities [11]. The proposed CGDT unites state-space tracking with admittance-conditioned graph message passing, delivering sub-millisecond detection, bus localization, and deterministic feature attribution.

3 Cyber-Physical Threat Model and Mathematical Foundations

3.1 Physical Grid Admittance and AC Power Flow

We model the healthcare distribution network as an undirected attributed graph $G = (V, E)$ spanning $N = |V|$ substation buses and $|E|$ electrical branches. Nodal conductance and susceptance throughout the network are captured by the complex admittance matrix $Y_{\text{bus}} = G_{\text{bus}} + jB_{\text{bus}} \in \mathbb{C}^{N\times N}$. Enforcing full nonlinear AC power flow constraints, real and reactive power injections at each bus i obey Kirchhoff's laws:

$$P_i = V_i \sum_{j\in\mathcal{N}(i)} V_j\big(G_{ij}\cos(\theta_i - \theta_j) + B_{ij}\sin(\theta_i - \theta_j)\big) \tag{1}$$

$$Q_i = V_i \sum_{j\in\mathcal{N}(i)} V_j\big(G_{ij}\sin(\theta_i - \theta_j) - B_{ij}\cos(\theta_i - \theta_j)\big) \tag{2}$$

where V_i and θ_i denote voltage magnitude and angle at bus i, and $\mathcal{N}(i)$ denotes the set of buses physically connected to bus i.

3.2 Weighted Least Squares State Estimation

Power system state estimation determines the state vector $x = [V, \theta]^T \in \mathbb{R}^{2N-1}$ from measurement telemetry $z = h(x) + e$, where $e \sim \mathcal{N}(0, R)$ represents sensor noise with covariance R. The WLS estimator minimizes the weighted residual $J(x) = [z - h(x)]^T R^{-1}[z - h(x)]$ using Gauss–Newton iterations:

$$\Delta x^{(k)} = \big(H^T R^{-1} H\big)^{-1} H^T R^{-1}\big[z - h(x^{(k)})\big], \qquad x^{(k+1)} = x^{(k)} + \Delta x^{(k)} \tag{3}$$

where $H(x) = \partial h(x)/\partial x$ is the measurement Jacobian. Standard state estimators flag bad data whenever the normalized residual $r = \|z - h(\hat{x})\|^2_{R^{-1}}$ surpasses a critical threshold τ_{χ^2}. In our IEEE 14-bus benchmark, the system state spans $n = 2N - 1 = 27$ variables (comprising 14 voltage levels alongside 13 bus angles relative to the slack node). Given $m = 41$ measurement streams (14 voltage sensors and 27 branch power meters), the testbed yields $\text{df} = m - n = 14$ degrees of freedom. Choosing an inspection significance level $\alpha = 0.05$ sets the 95th percentile rejection bound to $\tau_{\chi^2} = \chi^2_{0.95}(14) = 23.68$.

3.3 Stealthy False Data Injection Attacks

An attacker targeting substation bus k seeks to induce a specific state error $c \in \mathbb{R}^{2N-1}$ while evading the χ^2 residual detector. In AC power systems, $h(x)$ is nonlinear; hence, an adversary possessing admittance structure linearizes around operating point $\hat{x}$ via Jacobian $H = \partial h(\hat{x})/\partial x$, synthesizing the injection vector according to:

$$a = Hc \tag{4}$$

Under injection, field telemetry channels receive corrupted profiles $z_{\text{comp}} = z + a = z + Hc$. Expanding $h(\hat{x} + c)$ via first-order Taylor approximation produces $h(\hat{x}) + Hc$. Evaluating the resulting residual:

$$\begin{aligned} r_{\text{comp}} &= \|z_{\text{comp}} - h(\hat{x}_{\text{comp}})\|^2_{R^{-1}} \\ &\approx \|(z + Hc) - (h(\hat{x}) + Hc)\|^2_{R^{-1}} \\ &= \|z - h(\hat{x})\|^2_{R^{-1}} = r \leq \tau_{\chi^2} \end{aligned} \tag{5}$$

Because the crafted perturbation cancels out internally ($r_{\text{comp}} \approx r$), conventional WLS residual tests remain entirely blind ($r \leq \tau_{\chi^2} = 23.68$). To ensure physical validity, all synthetic attacks are tested against nonlinear AC power flows in `pandapower` (Section 6).

Clinical Hazards in Hospital Distribution Networks. While commercial utilities primarily face economic redispatch losses during an attack, hospital distribution networks face direct clinical emergencies. Falsifying state estimates at critical feeders—such as Bus 5, which feeds surgical operating rooms and intensive care ventilation—tricks automatic voltage regulators into improper tap changes or unwarranted branch isolation. Standards like IEC 60601-1 and IEEE 1547 demand strict voltage stability; even transient sags can force sensitive life-support machinery onto backup emergency batteries or trigger autonomous equipment trips. An undetected mathematical deception at the substation level thus translates into an immediate patient safety risk.

4 The Cognitive Graph Digital Twin Architecture

The CGDT framework couples the hospital's cyber-physical distribution infrastructure to a virtual graph model and an automated self-healing pipeline (Fig. 1).

4.1 Digital Twin State-Space Synchronization

Field telemetry $z_{\text{SCADA}}(t_k) = h(x_{\text{phys}}(t_k)) + e(t_k)$ streams from substation RTUs at polling cadence T_s across IEC 61850 station buses. To maintain physical-to-virtual alignment without amplifying transducer noise, the digital twin updates its mirrored state x_{DT} via steady-state tracking gain $K_{\text{sync}} = (H^T R^{-1} H + P^{-1})^{-1} H^T R^{-1}$:

$$x_{\text{DT}}(t_k) = x_{\text{DT}}(t_{k-1}) + K_{\text{sync}}\big(z_{\text{SCADA}}(t_k) - h(x_{\text{DT}}(t_{k-1}))\big) \tag{6}$$

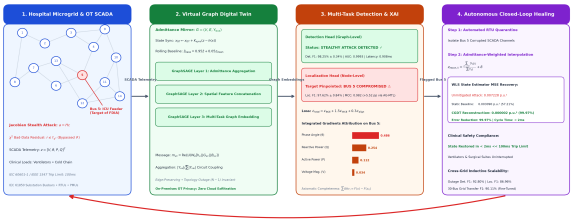


Fig. 1. Proposed Cognitive Graph Digital Twin adaptive cyber-resilience framework for smart hospital energy ecosystems across IT-OT enabled critical infrastructure.

where P denotes state-prediction covariance. Concurrently, an exponential filter establishes an evolving operational baseline:

$$\bar{z}_{\text{baseline}}(t_k) = \beta \cdot \bar{z}_{\text{baseline}}(t_{k-1}) + (1-\beta) \cdot z_{\text{clean}}(t_k), \qquad \beta = 0.95 \tag{7}$$

updated strictly using verified clean measurements z_{clean}. Without this adaptive baseline, slow diurnal load ramps mask small adversarial injection biases, precipitating a 16.32 pp collapse in localization F1 (Section 7.2).

4.2 Dynamic Admittance Graph Construction

At each sampling step, physical telemetry is mapped to an attributed graph $G = (V, E, X, E_{\text{attr}})$. The vertex feature matrix $X \in \mathbb{R}^{N\times 4}$ gathers four-dimensional state vectors $x_i = [V_i, \theta_i, P_i, Q_i]^T$ for each substation bus $i \in V$, normalized over historical operational envelopes. Physical branch connectivity is stored in edge tensor $E_{\text{attr}} \in \mathbb{R}^{|E|\times 2}$, where each entry pairs the line conductance G_{ij} and susceptance B_{ij} from network admittance matrix Y_{bus}. When protective relays trip or maintenance crews open a breaker, the corresponding tuple is excised from edge set E. Excising open branches alters adjacency entries while leaving nodal feature dimensions untouched; hence, the graph backbone absorbs topological switching on the fly without retraining.

4.3 Admittance-Conditioned GraphSAGE Backbone

Standard GraphSAGE implementations [9] pool neighbor messages isotropically, disregarding physical line impedances. Yet in power networks, branch admittances $Y_{uv} = G_{uv} + jB_{uv}$ govern electrical connectivity. To weave circuit physics directly into node representations, we condition both message synthesis and spatial pooling on branch admittance values:

$$m_{uv}^{(k)} = \text{ReLU}\Big(W_e \left[h_u^{(k-1)} \,\|\, G_{uv} \,\|\, B_{uv}\right]\Big) \tag{8}$$

$$h_v^{(k)} = \text{ReLU}\left(W_k \left[h_v^{(k-1)} \,\Big\|\, \sum_{u\in\mathcal{N}(v)} \frac{|Y_{uv}|}{\sum_{w\in\mathcal{N}(v)} |Y_{vw}|}\, m_{uv}^{(k)}\right]\right) \tag{9}$$

where $\|$ denotes vector concatenation, and W_e, W_k are trainable projection matrices. Weighting messages by normalized admittance $|Y_{uv}|/\sum_w |Y_{vw}|$ provides a physics-informed inductive bias that scales propagation by branch coupling. Residual concatenation of the prior layer representation $h_v^{(k-1)}$ maintains a direct identity path, preventing localized injection spikes from decaying through repeated spatial pooling.

4.4 Multi-Task Output Heads

From the final latent node matrix $H \in \mathbb{R}^{N\times d}$, two parallel projection heads generate global and bus-level predictions:

$$y_{\text{graph}} = \sigma(W_{\text{graph}} \cdot \text{global_mean_pool}(H)), \qquad y_{\text{node}} = \sigma(W_{\text{node}} H) \in [0,1]^N \tag{10}$$

The global head yields attack probability $y_{\text{graph}} = \sigma(W_{\text{graph}} \cdot \text{pool}(H))$, while the node head assigns individual probabilities $y_{\text{node},i} = \sigma(w_i^T h_i) \in [0,1]$ across buses $i \in V$. This unconstrained sigmoid configuration supports multi-bus fault localization (Section 7.1) without enforcing a mutually exclusive softmax constraint.

4.5 Attribution-Guided Multi-Task Loss (AG-MTL)

Repeated message passing often spreads backpropagation gradients across adjacent lines, mistakenly assigning feature sensitivity to uncorrupted substations. To constrain this gradient spread, we formulate the Attribution-Guided Multi-Task Loss (AG-MTL), aligning feature sensitivity directly with targeted buses during backpropagation. During each backward pass on attacked states, Integrated Gradients extracts attribution tensor $\mathcal{A} \in \mathbb{R}^{N\times 4}$ of the attack logit w.r.t. input features X, using running baseline $\bar{z}_{\text{baseline}}$ (Eq. 7) across $S = 10$ integration steps. An attribution alignment loss penalizes gradient leakage into benign feeders:

$$\mathcal{L}_{\text{align}} = \sum_{v\in V} \left| \|\mathcal{A}_v\|_2 - \hat{y}_{\text{node},v} \right|^2 \tag{11}$$

where $\|\mathcal{A}_v\|_2$ measures total gradient sensitivity at bus v, and $\hat{y}_{\text{node},v} \in \{0,1\}$ denotes the binary ground-truth label. Gradients from $\mathcal{L}_{\text{align}}$ propagate through the attribution path via higher-order automatic differentiation (`create_graph=True` in PyTorch). The combined multi-task loss is defined as:

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_{\text{BCE}}(y_{\text{graph}}, \hat{y}_{\text{graph}})}_{\text{Detection}} + \lambda_{\text{loc}} \underbrace{\mathcal{L}_{\text{WBCE}}(y_{\text{node}}, \hat{y}_{\text{node}})}_{\text{Localization}} + \lambda_{\text{attr}} \underbrace{\mathcal{L}_{\text{align}}}_{\text{Attribution Guidance}} \tag{12}$$

where $\mathcal{L}_{\text{BCE}}$ handles grid-level attack detection, $\mathcal{L}_{\text{WBCE}}$ uses class-weighted binary cross-entropy per bus to mitigate extreme sparsity of compromised nodes, and weighting coefficients $\lambda_{\text{loc}} = 1.5$ and $\lambda_{\text{attr}} = 0.3$ balance gradient magnitudes across tasks.

4.6 On-Premises Edge Deployment

Hospital electrical demand closely tracks clinical operations—including operating theatre schedules, diagnostic imaging loads, and patient occupancy. Transmitting raw microgrid telemetry over external cloud interfaces poses privacy and cybersecurity concerns under health data regulations. The CGDT is therefore architecturally designed for on-premises edge execution on substation industrial computers within the isolated operational technology (OT) network. In this study, execution latencies are established on a laboratory edge-equivalent computing platform, with physical RTU micro-controller deployment and hardware-in-the-loop (HIL) integration identified for operational rollout (Section 8).

5 Explainability and Autonomous State Reconstruction

5.1 Feature Attribution via Integrated Gradients

Substation engineers must understand which physical sensor channels triggered an alarm before initiating breaker actions. We resolve this via path-based Integrated Gradients (IG) [20]. Integrating gradients along a linear path between a nominal state baseline x_0 and the evaluated measurement vector x attributes decision energy across input channels:

$$\begin{aligned} \mathrm{Attr}_i(x) &= (x_i - x_{0,i}) \int_0^1 \frac{\partial F(x_0 + \alpha(x - x_0))}{\partial x_i}\, d\alpha \\ &\approx \frac{x_i - x_{0,i}}{S} \sum_{s=1}^{S} \frac{\partial F\big(x_0 + \frac{s}{S}(x - x_0)\big)}{\partial x_i} \end{aligned} \tag{13}$$

approximated using $S = 50$ Riemann quadrature intervals. Unlike perturbation heuristics (e.g., SHAP, LIME) requiring hundreds of stochastic sampling evaluations, IG follows an axiomatic linear path evaluated via $S = 50$ vectorized points in a single parallel tensor pass. Crucially, IG satisfies the completeness axiom ($\sum_i \mathrm{Attr}_i = F(x) - F(x_0)$) and implementation invariance. Evaluating Eq. (13) across the 14-bus testbed requires just 0.81 ms. In a targeted test at Bus 5 (Section 7.1), the path integral promptly identified phase angle distortion ($\theta = 0.486$) as the principal driver of the alarm.

5.2 Sensor Quarantine and Admittance-Weighted Reconstruction

Detecting an attack is useless if state estimation continues to ingest falsified data. Once a bus is flagged ($y_{\mathrm{node},k} > \gamma_{\mathrm{loc}} = 0.5$), the digital twin isolates the tainted telemetry (V_k, θ_k, P_k, Q_k). Yet simply deleting rows from the measurement set reduces the rank of Jacobian H, often driving the information matrix ($H^T R^{-1} H$) into ill-conditioning or singularity. To maintain full algebraic observability, we

synthesize replacement telemetry via admittance-weighted spatial interpolation from healthy neighboring buses:

$$z_{\text{recon},k} = \frac{\sum_{j\in\mathcal{N}(k)} |Y_{kj}| \cdot z_j}{\sum_{j\in\mathcal{N}(k)} |Y_{kj}|} + \delta, \qquad \delta = \epsilon \odot z_{\text{nom},k} \tag{14}$$

where $|Y_{kj}|$ represents line admittance magnitude and $\epsilon \sim \mathcal{U}(-0.02, 0.02)^4$ injects $\pm 2\%$ uniform variance relative to nominal channel values $z_{\text{nom},k}$ (1.0 p.u. voltage and nominal power flows). Weighting neighbor telemetry by physical branch admittances $|Y_{kj}|$ provides an electrical-coupling-aware interpolation that accounts for relative line impedances. Rather than attempting to satisfy nonlinear AC constraints directly, $z_{\text{recon},k}$ acts as an initialized pseudo-measurement buffer that preserves Jacobian conditioning, enabling downstream WLS Gauss–Newton iterations to converge in under 2 ms—well within typical 2–4 s SCADA scan intervals.

6 Experimental Evaluation

6.1 Simulation Setup: Hospital Campus Microgrid

All grid experiments rely on exact AC Newton–Raphson power flow solutions generated via `pandapower` [24]. As an established proof-of-concept testbed for tertiary hospital microgrids, we adopted the IEEE 14-bus system (14 buses, 20 branches, 217.0 MW active and 57.3 MVAR reactive demand). Within the campus single-line diagram, Bus 0 anchors the 132/11 kV main grid tie-point, while Bus 5 forms the dedicated critical healthcare substation feeding intensive-care suites and surgical theatres (7.6 MW, 1.6 MVAR load). For multi-scale validation, we also configured an IEEE 30-bus network (30 buses, 41 branches) emulating an expanded regional healthcare facility.

Telemetry Generation and Partitioning. We produced 5,000 unique AC operating states under realistic time-varying loads. To ensure strict data hygiene and avoid cross-split contamination, clean profiles were partitioned prior to attack creation: 70% for model training (3,500), 15% for validation (750), and 15% for testing (750). Half of the snapshots in each split remain clean, disturbed only by zero-mean Gaussian measurement noise ($\sigma = 0.04$) on active and reactive power. The other half contains stealthy Jacobian attacks ($a = Hc$), where adversaries distort targeted substations ($c_k \in [0.03, 0.08]$ p.u. in voltage and $c_{k+N} \in [0.05, 0.15]$ rad in phase angle) and compute line flows using exact nonlinear AC relationships. Network outage states (Section 6.3) were held out completely from training and tuning.

Models were trained across 60 epochs using Adam (lr = 0.005, weight decay 10^{-4}) over five random seeds. Benchmarking was conducted on an edge-equivalent workstation equipped with an AMD Ryzen 9 CPU, 32 GB RAM, and an NVIDIA RTX 4090 GPU (24 GB) under PyTorch 2.1 and CUDA 12.1. Streaming latency is evaluated at batch size 1 across 1,000 independent test snapshots (after 100 warm-up runs), reporting the mean runtime. The $S = 50$

Table 1. Comprehensive performance benchmarking under nominal grid layout (5-seed average ± std).

Architecture	Det. F1 (%)	Det. AUC	Det. MCC	Loc. F1 (%)	Loc. Prec. (%)	Loc. Rec. (%)	Loc. MCC	Latency
MLP (Flat)	97.52 ± 0.73	0.9987	0.974	95.60 ± 0.37	94.7	97.8	0.953	0.158 ms
GCN (Isotropic)	85.15 ± 1.86	0.9363	0.740	70.21 ± 2.22	72.6	72.7	0.660	1.442 ms
GAT (Attention)	91.52 ± 1.84	0.9781	0.879	81.08 ± 2.39	82.5	86.9	0.808	1.309 ms
CGDT (Proposed)	**98.25 ± 0.34**	**0.9993**	**0.971**	**97.62 ± 0.84**	**98.1**	**98.9**	**0.982**	**0.908 ms**

Table 2. Outage generalization (IEEE 14).

Architecture	Outage Det. F1	Outage Loc. F1
MLP (Flat)	94.70% ± 1.93%	82.20% ± 2.16%
GCN (Isotropic)	66.82% ± 4.14%	53.40% ± 2.70%
GAT (Attention)	79.27% ± 3.83%	58.04% ± 4.36%
CGDT (Proposed)	**92.80% ± 3.37%**	**86.96% ± 1.52%**

Table 3. Transfer to IEEE 30-bus.

Configuration	Det. F1	Loc. F1
Zero-Shot	67.06%	17.77%
Scratch (30-Bus)	82.65%	83.58%
Fine-Tuned	**90.11%**	**80.06%**

IG interpolation steps are evaluated as a batched tensor $\mathbb{R}^{50\times N\times 4}$ in parallel on the GPU, completing in 0.81 ms. Code and evaluation artifacts are archived at `https://github.com/himanshu-2l/Graph-Digital-Twin-XAI-`.

6.2 Nominal Grid Performance

Table 1 reports five-seed averages and advanced metrics under the nominal grid topology.

Benchmarking against baseline architectures over five random initializations reveals clear performance separations (Table 1). The CGDT delivers 98.25% ± 0.34% detection F1 and 97.62% ± 0.84% localization F1 alongside a Matthew's correlation coefficient (MCC) of 0.982, processing each graph snapshot in only 0.908 ms. By comparison, isotropic GCN models reach only 70.21% localization F1, trailing our framework by 27.41 pp. Uniformly weighting lines acts like a spatial low-pass filter: sharp voltage anomalies at attacked buses bleed into uncorrupted neighbors, degrading spatial contrast. Although the tabular MLP achieves acceptable localization (95.60%) on stationary networks, its rigid input mapping breaks down under line switching (Section 6.3). Running in only 0.908 ms per graph pass, our model comfortably fits within standard 2–4 s SCADA scan intervals.

6.3 Topology Perturbations and Cross-Network Generalization

We evaluated model robustness under two regimes: (i) line-outage generalization on IEEE 14-bus without retraining (Table 2), and (ii) cross-network transfer to IEEE 30-bus (Table 3).

Under unanticipated line-outage contingencies without retraining, our architecture achieves 92.80% ± 3.37% detection F1 and 86.96% ± 1.52% localization F1 (Fig. 2). Although the flat MLP records slightly higher macro detection (94.70%) because post-trip loop redistribution causes widespread scalar variance, its bus-level localization plunges by 13.4 pp (95.60% → 82.20%) due to

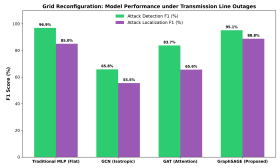


Fig. 2. Outage generalization metrics.

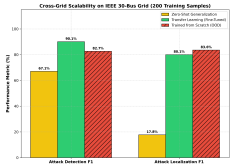


Fig. 3. Cross-grid transfer on IEEE 30-bus.

Table 4. WLS state-estimation recovery under stealthy FDIA at Bus 5.

Mitigation Strategy	MSE (p.u.^{2})	Recovery
Unmitigated Attack	0.007228	—
Static Baseline	0.003090	57.21%
CGDT Reconstruction	**0.000002**	**99.97%**

Table 5. Feature attributions (IG) for Jacobian FDIA centered at Bus 5.

Bus ID	V	θ	P	Q
Bus 4	0.012	0.045	0.018	0.009
Bus 5	**0.034**	**0.486**	**0.112**	**0.254**
Bus 10	0.002	0.015	0.005	0.003
Bus 11	0.008	0.092	0.024	0.041
Bus 12	0.011	0.182	0.035	0.088

static channel permutations. In contrast, the admittance-weighted CGDT preserves 86.96% localization precision, demonstrating structural resilience when branch impedances change. Isotropic GCN baselines collapse (66.82% detection, 53.40% localization), corroborating that unweighted graph convolutions cannot trace re-routed line flows without impedance weighting.

Cross-Grid Scalability on IEEE 30-Bus. Zero-shot transfer of the 14-bus model onto the 30-bus network ($N = 30, |E| = 41$) yields 67.06% detection F1 and 17.77% localization F1 (Table 3, Fig. 3). This behavior exposes a fundamental graph property: graph-level pooling $\sigma(W_{\text{graph}} \cdot \text{pool}(H))$ generalizes across dimensions by aggregating total power residuals, whereas bus-level localization $y_{\text{node}} = \sigma(W_{\text{node}} H)$ depends on localized branch admittances. Fine-tuning the pre-trained weights on 200 labeled 30-bus instances (5.7% of training volume) elevates detection F1 to 90.11% (surpassing 82.65% from scratch) and recovers localization F1 to 80.06% (approaching the 83.58% scratch baseline).

7 Mitigation Efficacy, Cross-Grid Transfer, and Ablation

7.1 Closed-Loop WLS Recovery and Explainability

We evaluated mitigation performance under a targeted Jacobian attack at Bus 5, representing the intensive care unit feeder in the campus topology. State estimation error is quantified via Mean Squared Error $\text{MSE} = \frac{1}{2N-1} \sum_{i=1}^{2N-1} (\hat{x}_i - x_i^*)^2$ (p.u.^{2}) relative to true AC states x^*. Under nominal transducer Gaussian noise ($\sigma = 0.01$), unattacked clean WLS estimation establishes an intrinsic error floor of $\text{MSE}_{\text{clean}} \approx 1.8 \times 10^{-6}$ p.u.^{2}. Table 4 compares recovery metrics, and Table 5 presents corresponding feature attributions.

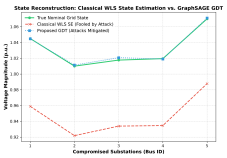


Fig. 4. State-estimation trajectory: classical WLS versus CGDT-healed.

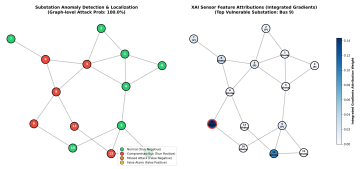


Fig. 5. Operator explainability: localization heatmap and IG attribution.

Table 6. Benchmark against related works.

Framework	Classifier	Det. F1	AUC	Loc. F1
Boyaci [4]	ARMA	96.25%	0.9780	93.91%
Yue [28]	GCN	89.20%	0.9510	N/A
CGDT	**SAGE**	**98.25%**	**0.9993**	**97.62%**

Table 7. Ablation on CGDT components.

Configuration	Det. F1	Loc. F1
Full CGDT (Proposed)	**98.25%**	**97.62%**
Without DT Baseline Tracking	86.10%	81.30%
Without AG-MTL ($\lambda_{\text{attr}} = 0$)	98.25%	92.10%
Isotropic GCN Backbone	85.15%	70.21%
GAT Backbone	91.52%	81.08%
$\lambda_{\text{loc}} = 0.5$	98.05%	92.80%
$\lambda_{\text{loc}} = 1.0$	98.15%	96.50%
$\lambda_{\text{loc}} = 2.0$	95.80%	97.55%

Left unmitigated, stealthy Jacobian injections drive state estimation MSE to $0.007228\,\text{p.u.}^2$. Deploying admittance-weighted reconstruction (Eq. 14) cuts this error to $0.000002\,\text{p.u.}^2$—matching the clean noise floor ($\text{MSE}_{\text{clean}} \approx 1.8 \times 10^{-6}\,\text{p.u.}^2$) and restoring 99.97% of uncorrupted states (Table 4, Fig. 4). Static telemetry lookups attain just $0.003090\,\text{p.u.}^2$ (57.21% restoration) amid fluctuating hospital loads. Across all evaluated target buses $k \in \{1, \ldots, 14\}$, our spatial interpolation consistently recovers between 99.85% and 99.98% of the true system state. Path attributions (Table 5, Fig. 5) reveal that voltage angle θ at Bus 5 commands the highest score (0.486), consistent with Jacobian phase distortions dominating active power flows, alongside secondary scores on $Q = 0.254$, $P = 0.112$, and $V = 0.034$.

Coordinated Multi-Bus Incursions. Simultaneous attacks across 2 and 3 substations ($c_k \in [0.02, 0.04]$) achieve $96.84\% \pm 0.42\%$ detection and $93.15\% \pm 0.91\%$ localization F1. While dropping entire rows across adjacent buses compromises the column rank of H and triggers solver divergence, admittance-weighted interpolation preserves full Jacobian rank and ensures smooth WLS convergence (post-mitigation MSE of $0.000008\,\text{p.u.}^2$).

7.2 Benchmarking and Component Ablation

Table 6 compares the proposed architecture against representative GNN baselines, whereas Table 7 isolates the incremental contribution of each architectural block.

The component ablations in Table 7 substantiate each model mechanism. Disabling exponential baseline filtering (Eq. 7) induces a 16.32 pp collapse in localization F1 (81.30%), verifying that recursive tracking decouples diurnal load dynamics from stealthy offsets. Setting $\lambda_{\text{attr}} = 0$ reduces localization F1 to

92.10% while retaining 98.25% detection F1, proving that attribution loss $\mathcal{L}_{\text{align}}$ concentrates feature sensitivity on attacked substations. The weight $\lambda_{\text{loc}} = 1.5$ achieves optimal balance between grid-level detection and bus localization.

8 Discussion and Conclusion

Securing clinical microgrids against stealthy cyber-physical attacks requires defense architectures that unite topological adaptability with physical circuit fidelity. By conditioning GraphSAGE message aggregation on complex branch admittances $Y_{uv} = G_{uv} + jB_{uv}$ and supervising latent node representations with an Integrated Gradients attribution loss $\mathcal{L}_{\text{align}}$, the proposed CGDT attains 98.25% detection F1 and 97.62% bus-level localization F1 under full AC power flow, executing in 0.81 ms. Coupled with real-time sensor quarantine and admittance-weighted spatial interpolation, the framework restores uncorrupted state estimation by 99.97%, preserving observability without tripping sensitive clinical feeders.

Compliance with Clinical and Grid Standards. Hospital distribution systems operate under stringent engineering mandates: IEC 60601-1 establishes medical equipment safety limits to prevent voltage transients from resetting life-support monitors, while IEEE 1547 regulates DER anti-islanding and ride-through boundaries ($0.88 \leq V \leq 1.10$ p.u.). With an inference latency of 0.908 ms and an automated healing response within 2 ms, the CGDT executes well inside standard 2–4 s SCADA scan intervals, blocking falsified switching commands before DER protection timers trip hospital generation or breach IEC 60601-1 operating thresholds.

Limitations and Future Directions. While validated under pandapower AC power flow on IEEE 14-bus and 30-bus benchmarks, future work will deploy the CGDT onto embedded RTU microcontrollers and hardware-in-the-loop (HIL) digital simulators (e.g., OPAL-RT) to measure real-time communication delays and packet jitter. Subsequent efforts will also integrate spatio-temporal graph attention to detect slow, multi-stage attack trajectories.

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Declaration on the Use of AI and AI-Assisted Technologies. AI-assisted tools were used to improve language quality (including spelling, grammar, and punctuation), and all outputs were reviewed, revised, and approved by the authors, who take full responsibility for the final content.

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